Design and Implementation of a Dual-Evaporator Transcritical High Temperature Heat Pump

Research Project

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1 Context

According to the Global Energy Review 2025 report published in March 2025 by the IEA [2] , the global CO_2 emissions have increased by 0.8% in 2024, reaching an all-time high of 37.8 Gt CO_2 a year. In addition, the report highlights that the energy demand grew by 2.2% in 2024. This increase in CO_2 emissions is totally inconsistent with the goals of the Paris Agreement, which aims to abate the CO_2 emissions by 43% by 2030 and to reach net-zero emissions by 2050. However, the lower increase of CO_2 emissions compared to the energy demand growth is an encouraging trend, as it suggests a relative decoupling of emissions from energy consumption.

In [5], we can read that the waste heat potential in the EU-27 and UK reached 221.32 TWh/year in 2021. In comparision, the demand for less than 140°C heat in Germany was 612 PJ, or 170 TWh in 2012 [4]. These numbers justify the interest from companies/researchers to design and study machines capable of recovering this waste energy.

In this work, we will focus on the use of high temperature heat pumps (HTHP's) for recovering waste heat from industrial processes. In [4], the authors provide a list of manufacturers and universities that have already been working on the design of several machines for a at least one decade. The already existing machines cover a large spectrum of applications, going from small machines of a few thousands watts to big ones of more than one megawatt. However, new design ideas can still be studied in order to improve the current recovery systems.

The main purpose of this work is to design the thermodynamical cycle of a transcritical dual-evaporator HTHP, build it and test its performances in laboratory conditions. No similar applications were found in the litterature. Beside, the machine build in the frame of this master thesis should be used in the future for an academic purpose.

2 Literature review

As discussed in the previous section, no publications specifically addressing dual-evaporator transcritical HTHPs were identified. Nevertheless, a brief review of the current state-of-the-art technology for HTHPs was carried out. This review has led to the identification of several reasons why the topic may be particularly relevant for certain applications and therefore is worth further investigation.

First, regarding the dual-evaporator configuration: it has been shown in [6] that using multiple evaporators offers significant potential for waste heat recovery applications. In the case study considered, notable improvements in both the Coefficient of Performance (COP) and Volumetric Heating Capacity (VHC) were observed when two evaporators were employed instead of one, once a significant portion of the available heat was present at the medium temperature (MT) level (see Figure 1).

This proportion of heat available at the medium temperature level relative to the total heat input is quantified by the heat source ratio parameter (β), first introduced in [3]:

$$\beta \triangleq \frac{\dot{Q}_{MT}}{\dot{Q}_{MT} + \dot{Q}_{LT}}$$

where $\dot{Q}_{MT}$ and $\dot{Q}_{LT}$ represent the thermal powers captured at the medium and low temperature evaporators, respectively.

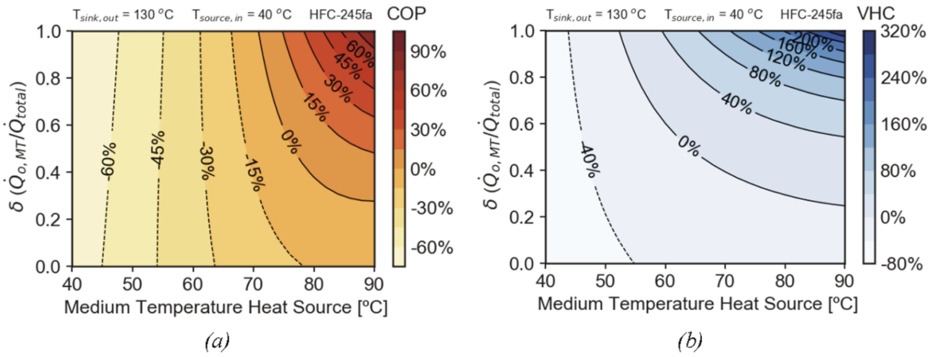

Fig. 15. Performance parameters evaluation of the TS Booster cycle compared to the reference TS Economizer, operating with HFC-245fa: a) COP and b) VHC.

Figure 1: COP and VHC increase significantly when using two evaporators instead of one, once a substantial portion of the total captured heat is available at the medium-temperature source. Figure taken from [6].

While the benefit of a higher COP is not to be demonstrated, the VHC is also of particular interest. As explained in [4], the VHC is directly related to the refrigerant charge and size of the compressor: a higher VHC allows for a smaller refrigerant charge and a more compact compressor, thereby reducing system costs. Consequently, both the COP and the VHC are important factors in designing efficient and cost-effective heat pumps.

Next, regarding the transcritical aspect of the cycle: in [7], the authors demonstrate that using a transcritical cycle instead of a subcritical one can lead to significant improvements in COP when the temperature glide at the heat sink is high. According to their study, the ideal heat sink temperature glide is around 60 K. This COP increase is attributed to a lower temperature mismatch during heat transfer at the heat sink (see Figure 2).

The improved temperature matching results from the properties of a supercritical fluid. Since heat transfer occurs above the critical pressure, the fluid does not undergo a phase change. Therefore, its temperature can vary significantly during heat transfer, allowing for better alignment with the heat sink fluid temperature and reduced exergy losses. The authors also show that transcritical cycles generally achieve higher VHC values than subcritical cycles.

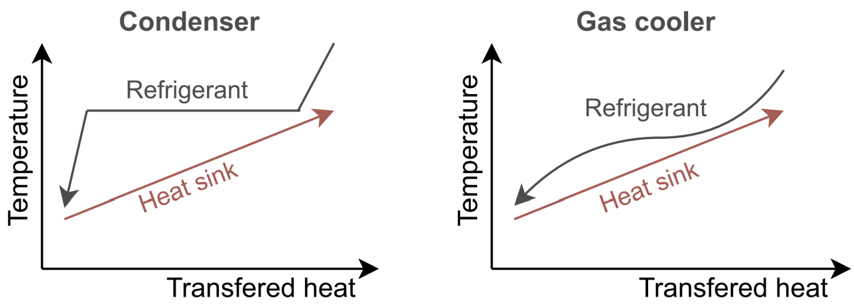

Fig. 1. Temperature profile during heat rejection for subcritical operation (condenser) and transcritical or supercritical operation (gas cooler).

Figure 2: The area between the two curves represents the exergy destruction due to heat transfer. This area is considerably smaller in the transcritical case, resulting in lower irreversibilities and a potentially higher COP. Figure taken from [7].

Finally, regarding high-temperature applications: heat pumps are classified as "high temperature" when the heat sink temperature exceeds 100 °C [4]. While conventional heat pumps are already widely used for applications below 100 °C, HTHPs are designed to provide process heat in the range of 100-200°C. According to the IEA, almost 40% of industrial heat demand is at levels suitable for heat pumps, with temperatures between 100°C and 200°C representing nearly half of this share (see Figure 3) [1].

Figure 1.16 ▷ Global industrial process heat demand by temperature in the APS, 2021 and 2030

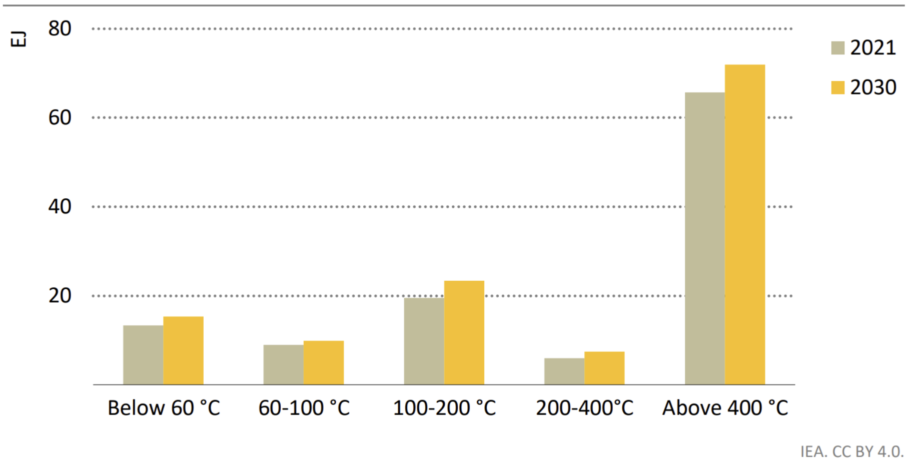

Figure 3: Around 22 EJ of industrial heat demand will be required in the range of 100-200 °C, corresponding to just under 20% of the expected total industrial heat demand in 2030. Figure taken from [1].

Today, most of this industrial heat is based on fossil fuels. Developing efficient high-temperature heat pumps is therefore a crucial step toward decarbonizing this 22 EJ of industrial heat demand between 100°C and 200°C, and more broadly, supporting the energy transition.

In conclusion, a dual-evaporator transcritical high-temperature heat pump has the potential to outperform current HTHPs on the market, particularly in applications where heat is available at two temperature levels (e.g., waste heat recovery) and where the temperature glide at the heat sink is high. The aim of this Master's thesis is to investigate this concept, contributing to the broader effort to advance high-temperature heat pumps and decarbonize industrial heat production.

3 Research question

As detailed previously, multiple studies on the efficiency of refrigeration cycles have been conducted in the recent years, compared to HTHP cycles. Our objective is to implement new design ideas, inspired from these studies like the dual evaporator or the transcritical state, to improve the efficiency of HTHP cycles.

This paper is adressed to the scientific community, but also to industrials since it aims to provide insights and solutions that can be directly applied in real-world applications. In addition, a prototype will be developed to validate the proposed concepts.

Our work is also multidisciplinary, as it will include thermodynamics, heat transfer, fluid mechanics, control techniques and machine design. It is a complete project, from the numerical design to the building and testing of a prototype.

The research question that will be adressed in this master thesis is the following:

How do the integration of a dual evaporator and the operation in a transcritical state influence the performance of a high-temperature heat pump cycle under real-life application conditions?

4 Project description

In order to answer to the previous research question, we distinguish six steps in this master thesis. These steps can be observed in the Figure ...

Preliminary thermodynamical cycle design

In this first part, we will numerically modeled the thermodynamical design. Each student will work simultaneously on different blocks, determined in advance. The analysis of the cycle will include an energy and exergy analysis to ensure optimal performances. The numerical model will be writen in Python and the module Coolprop will be used to access numerical thermodynamics tables.

Final cycle design

Assembling all the different blocks will be the purpose of the second part. Beside connecting the blocks, an optimization process will be developped to determine the optimal parameters in nominal conditions. In addition, more physical aspects will be added to the model such as head losses computations and heat transfer modeling.

Control

One main advantage of this machine will be its capacity of simulating different conditions thanks to the dual evaporator. A control logic must therefore be implemeted. This part is critical because we'll have to think about the nature and the location of the sensors. A student in industrial engineering will bring his expertise and will start to collaborate with us.

Machine design

In this part, the other student will be in charge of the design of the machine. It will contain material/components selection, heat exchangers and insulation design and the elaboration of precise technical draw. That will be our task to correct the final cycle design (head losses, heat exchange, ...). We will also collaborate with Frank, the technician that will be responsible of the buiding of the phase. We surely think that its expertise will be needed for many design uncertainties.

Building

The building phase will be ensured by Frank and other technicians of the CREDEM.

Testing

Finally, the last part will be dedicated to the evaluation of the machine. Several tests will take place to ensure the good operation of the machine and to collect data in order to compare them to theoritical values. We don't think it will possible to have a correction phase in case of underperforming.

5 Share of workload and responsibilities

Taking the different steps of the project presented in the previous section, the workload will be shared as follows:

1. **P&ID and Preliminary Thermodynamic Design:** The EPL students will work in parallel on different sub-parts of the `Python` code, but strategic design decisions will be made jointly and mutual support will be provided as needed.

2. **Optimization of the Thermodynamic Design:** The optimization stategy will be jointly developed by the EPL students to maximize the performance of the system in nominal conditions.

3. **Sizing of the Main Components:** Similar to the first phase, the EPL students will work in parallel on different components to maximize productivity, as this phase must be completed by the end of the first semester to ensure availability of the components for the prototyping phase.

4. **Establishment of a First Control Strategy:** All students will participate in the development of a first control strategy for the prototype.

5. **CAD and Operational Model:** The ECAM student will lead the CAD work, with technical support from a CREDEM platform technician. The resulting Operational Model will be validated by the EPL students based on their initial thermodynamic design.

6. **Prototype Construction and Instrumentation:** The prototype will be built under the supervision of a CREDEM technician.

7. **Safety Validation of the Prototype:** This step will be fully managed by the CREDEM platform.

8. **Control Logic and Performance Validation:** Once the prototype is operational, the students will carry out the validation of control logic and performance, as well as fine-tune the prototype to optimize its operation.

Throughout the project, weekly meetings will be organized between the students to discuss progress, share knowledge, and make important decisions. In addition, students will meet regularly with their mentors to present key results, receive feedback, and discuss challenges. Progress will also be reported formally during monthly meetings with the supervisor.

6 Roadmap and planning

To be completed by HW (2 pages).

7 Project team

The project team is composed of the following members:

- **EPL students:** Warscotte Hugo, Grégoire Franklin

- **ECAM student:** To be determined
- **CREDEM platform technician:** Hesbois Frank
- **Mentors:** Hauglustaine Mattéo, Laterre Antoine
- **Supervisor:** Prof. Contino Francesco

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Note on AI assistance

This report was proofread with the assistance of `ChatGPT`, an AI language model developed by `OpenAI`.